Probability Theory and Random Processes (MA225)

Lecture Slides

Lecture 21

Stochastic Processes

Def: Let $T = \{0, 1, 2, \ldots\}$ or $T = [0, \infty)$. A stochastic process $\{X_t\}_{t \in T}$ is a collection of real valued random variables.

- That means for each $t \in T$, X_t is a random variable.
- We will think of t as a time and call X_t the state of the process at time t.
- When $T = \{0, 1, 2, \ldots\}$, we call $\{X_t\}$ is called a discrete time stochastic process.
- For $T = [0, \infty)$, $\{X_t\}$ is called continuous time stochastic process.

Example

Example 1: Price of gold on each day

Example 2: Minimum temperature of each day

Example 3: Total number of visitors in an amusement park up to time t of a day.

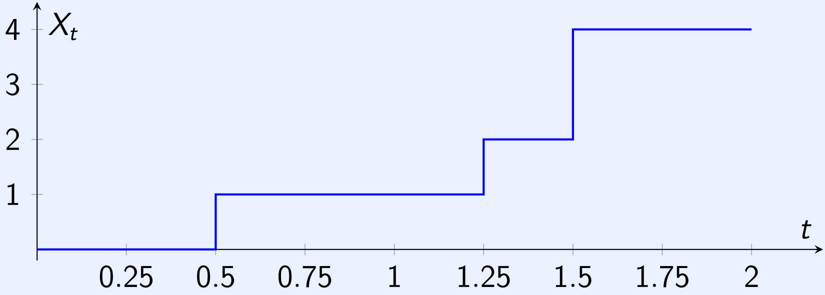
Terminologies

- The set of all possible values taken by a stochastic process $\{X_t\}$ is called **state space**.
- We will only consider atmost countable state spaces.
- X_t is a discrete random variable for all t.
- The PMF of X_0 is called the **initial distribution**.
- Any realization of $\{X_t\}$ is called a **sample path**.

Example

Example 4: A sample path for Example 3 is given below.

- Suppose that the park opens at 9 am ($t = 0$).
- The first visitor arrives at 9:30 am ($t = 0.5$)
- The second visitor arrives at 10:15 am ($t = 1.25$)
- Two more visitors arrive at 10:30 am ($t = 1.5$)
- So on . . ..



Example

Example 5: Consider throwing a fair die. Let $X_0 = 0$ and X_n denote the outcome of the nth throw, $n \geq 1$. Then, $\{X_n\}_{n \geq 0}$ is a stochastic process.

- The state space is $\{0, 1, 2, \ldots, 6\}$.
- X_n's are independent.
- The initial distribution is δ_0.
- For $n \geq 1$, $P(X_n = k) = \frac{1}{6}$, $k = 1, 2, \ldots, 6$.

Example

Example 6: Consider throwing a fair die. Let $X_0 = 0$ and X_n denote the number of sixes in the first n throws, $n \geq 1$. Then, $\{X_n\}_{n \geq 0}$ is a stochastic process.

- The state space is $S = \{0, 1, 2, \ldots\}$.
- X_n's are dependent.
- The initial distribution is δ_0.
- For $n \geq 1$ and $i \in S$,

$$P(X_n = j | X_{n-1} = i) = \begin{cases} \frac{1}{6} & \text{for } j = i + 1 \\ \frac{5}{6} & \text{for } j = i \\ 0 & \text{otherwise.} \end{cases}$$

Markov Chain

Def: A stochastic process $\{X_n\}_{n\geq 0}$ is said to be a Markov chain (MC) if

$$P\left(X_{n+1}=j|X_n=i,\, X_{n-1}=i_{n-1},\, \ldots,\, X_0=i_0\right)=P\left(X_{n+1}=j|X_n=i\right)$$

for all states $i_0,\, i_1,\, \ldots,\, i_{n-1},\, i$ and j and all $n\geq 0$.

Remark: For a MC, the conditional distribution of X_{n+1} (future) given the past $X_0,\, X_1,\, \ldots,\, X_{n-1}$ and present X_n depends only on the present X_n and not on past $X_0,\, X_1,\, \ldots,\, X_{n-1}$.

Time-homogeneous Markov Chain

Def: A MC $\{X_n : n \geq 0\}$ is said to be a time-homogeneous MC if

$$P\left(X_{n+1} = j | X_n = i\right) = P\left(X_n = j | X_{n-1} = i\right) = p_{ij}$$

for all $n \geq 1$.

- We will only consider time-homogeneous MC in this course.
- $p_{ij} \geq 0$ for all i and j.
- $\sum_{j=0}^{\infty} p_{ij} = 1$.
- p_{ij} is called one-step transition probability from state i to state j.
- The matrix $P = \left(p_{ij}\right)_{i,j \geq 0}$ is called one-step transition probability matrix (TPM).

Example

Example 7: Suppose that the chance of rain tomorrow depends on previous weather conditions only through whether or not it is raining today and not on past weather condition. Suppose that if it is raining today, then it will rain tomorrow with probability α. If it is not raining today, then it will rain tomorrow with probability β.

- Let 0 denote the state that it is not raining and 1 denote the state that it is raining.
- $X_n =$ State of the nth day.
- State space is $\{0, 1\}$.
- The one-step transition probability matrix is

$$\begin{pmatrix} 1-\beta & \beta \\ 1-\alpha & \alpha \end{pmatrix}.$$

Example

Example 8: Suppose that the chance of rain tomorrow depends on previous weather conditions through last two days. Assume

$$P\left(\text{Rain tomorrow}\middle|\text{rain for past two days}\right) = 0.7$$

$$P\left(\text{Rain tomorrow}\middle|\text{rain today, no rain yesterday}\right) = 0.5$$

$$P\left(\text{Rain tomorrow}\middle|\text{rain yesterday, no rain today}\right) = 0.4$$

$$P\left(\text{Rain tomorrow}\middle|\text{no rain for past two days}\right) = 0.2$$

- If X_n is the state of the nth day, then $\{X_n : n \geq 0\}$ is not a Markov Chain.

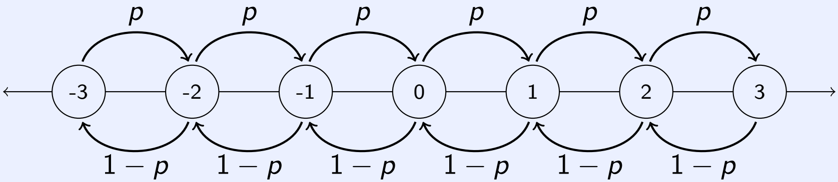
Example

- Let use define the states as
  - 0 if rain both today and yesterday
  - 1 if rain today and no rain yesterday
  - 2 if no rain today and rain yesterday
  - 3 if no rain both today and yesterday
- If Y_n is the state (as defined above) on the nth day, then $\{Y_n\}$ is a MC with TPM

$$\begin{pmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{pmatrix}.$$

Example

Example 9: A MC whose state space is given by set of integers is said to be simple random walk (SRW) if for some $0 < p < 1$, $p_{i,i+1} = p = 1 - p_{i,i-1}$. It is said to be simple symmetric random walk (SSRW) if $p = 1/2$.



Example

Example 10: [A Gambling Model] Consider a gambler who at each play of the game either wins Re. 1 with probability p or losses Re. 1 with probability $1 - p$. Suppose that the gambler quits plays either when he goes broke or he attains a fortune of Rs. N.

- Let X_n denote the fortune of the gambler after the nth game.
- $\{X_n : n \geq 0\}$ is a MC with with state space $\{0, 1, \ldots, N\}$.
- The one-step transition probabilities are

$$
\begin{aligned}
& p_{00} = p_{NN} = 1 \\
& p_{i,i+1} = p, \; p_{i,i-1} = 1 - p \quad \text{for} \quad i = 1, 2, \ldots, N - 1 \\
& p_{ij} = 0 \quad \text{otherwise.}
\end{aligned}
$$